



Ask

WEBHOOKS



Ask



skills-activity-ended

Event triggered when a learner completes or exits a training activity

Event Overview

The `skills-activity-ended` event is emitted when a learner completes, times out of, or intentionally exits a training activity in Skills Software. It includes final scoring data, performance metrics, and the learner's completion status so downstream systems can close out the attempt. Most attempts begin with a matching `skills-activity-started` event, but an end event may be absent if the learner abandons the session unexpectedly.

Delivery Behavior

- Fires once an attempt concludes because the learner finished, failed, or the scenario timed out due to inactivity.
- Not delivered if the browser closes or the learner loses connectivity before the Skills Software can finish the attempt; in that case only the start event is sent.
- When a learner does nothing, the platform ends the attempt automatically and emits this event with a timeout-related `sessionOutcome`.
- `activity_elapsed` captures the full duration of the attempt, including pauses or idle time recorded before completion.

- Subsequent retries of the same `activity_id` emit new start events and, when the attempt concludes cleanly, a corresponding end event with updated failure counts and detailed results.

Event Type

`skills-activity-ended`

Webhook Payload

```
{
  "requestId": "548994-1",
  "eventType": "skills-activity-ended",
  "timestamp": "2026-01-13T07:56:24.5534575Z",
  "stationId": "696224442a77d0120654fbab",
  "data": {
    "launch_id": "548994-1",
    "event_type": "activity-ended",
    "org_id": "10001234",
    "manikin_id": "17700000000",
    "user_email": "user.name@example.com",
    "user_id": "12345",
    "user_name": "User Name",
    "activity_id": "skills-web-adult-compressions",
    "activity_completed": false,
    "activity_score": 0,
    "activity_elapsed": 82280,
    "activity_failures": 1,
    "organization_name": "Distributor A",
    "institution_id": "dfcfb5bc-dd3a-5201-8e17-040e33da9b73",
    "institution_name": "Institution A",
    "time_zone": "Europe/Dublin",
    "activity_result": {
      "SSJVersion": "1.21",
      "CprSessionStatistics": {
        "compCount": 60,
        "compCyclesCount": 1,
        "ventCount": 0,
        "ventCyclesCount": 0,
        "compInactivityPeriodCount": 0,
        "compMeanDepth": 49,
        "compStdDeviationDepth": 58,
        "compMeanReleaseDepth": 0,
        "compStdDeviationReleaseDepth": 0,
        "compMeanRate": 100,
        "compStdDeviationRate": 61,
        "compCorrectDepthPercentage": 60,
        "compCorrectReleasePercentage": 100,
        "compCorrectRatePercentage": 41,
        "compTooLowRatePercentage": 59,
        "compHandPosCorrectPercentage": 22,
```

```

    "inactivityAverageTime": 0,
    "meanVentilationInflationTime": null,
    "stdDevVentilationInflationTime": null,
    "numPreVents": null,
    "averageNumberOfVentilationsPerCycle": null,
    "minNumberOfVentilationsPerCycle": null,
    "maxNumberOfVentilationsPerCycle": null,
    "averageNumberOfCompressionsPerCycle": null,
    "minNumberOfCompressionsPerCycle": null,
    "maxNumberOfCompressionsPerCycle": null,
    "ventMeanVolume": null,
    "ventStdDeviationVolume": null,
    "ventMeanRate": null,
    "ventStdDeviationRate": null,
    "ventCorrectVolumePercentage": null,
    "ventNotEnoughVolumePercentage": null,
    "ventTooMuchVolumePercentage": null,
    "ventCorrectRatePercentage": null,
    "ventTooLowRatePercentage": null,
    "ventCorrectInflationTimePercentage": null,
    "ventTooFastInflationTimePercentage": null,
    "cprCyclesAccepted": null,
    "flowTimePercentage": 100
  },
  "CprSessionScores": {
    "cprScore": 8,
    "overallScore": 8,
    "categoryThresholdBasic": 75,
    "categoryThresholdAdvanced": 75,
    "compReleaseScoreReduction": 0,
    "compDepthShallowScoreReduction": 23,
    "compRateLowScoreReduction": 4,
    "compRateHighScoreReduction": 0,
    "compHandposScoreReduction": 89,
    "compOverallScore": 8,
    "ventShallowScoreReduction": 0,
    "ventExcessiveScoreReduction": 0,
    "ventRateLowScoreReduction": 0,
    "ventRateHighScoreReduction": 0,
    "ventCyclesScoreReduction": 0,
    "ventOverallScore": null,
    "compDeepDepthScoreReduction": 0,
    "compCyclesScoreReduction": 0,
    "compFlowtimeScoreReduction": 0,
    "preVentsTooFewScoreReduction": 0,
    "preVentsTooManyScoreReduction": 0,
    "inflationTimeFastScoreReduction": 0,
    "inflationTimeSlowScoreReduction": 0,
  }
}

```

```

    "isScoringModeAED": false,
    "compFlowTimeScore": 100,
    "compDepthScore": 77,
    "compRateScore": 96,
    "compReleaseScore": 100,
    "compHandPositionScore": 11,
    "ventRateScore": null,
    "ventVolumeScore": null,
    "aedScore": null,
    "aedAppliedPadsScore": null,
    "aedShockDeliveredScore": null,
    "aedCprStartedScore": null,
    "aedCprAfterNoShockScore": null,
    "virtualRescuerInterferenceScoreReduction": null,
    "aedInterferenceScoreReduction": null,
    "aedMissingAnalysisStayClearScoreReduction": null,
    "aedMissingShockStayClearScoreReduction": null,
    "compressionsPerSetManyScoreReduction": null,
    "compressionsPerSetFewScoreReduction": null,
    "ventilationsPerSetManyScoreReduction": null,
    "ventilationsPerSetFewScoreReduction": null,
    "scanChestTimeSlowScoreReduction": null,
    "scanChestTimeFastScoreReduction": null
  },
  "DeviceInfo": {
    "serialNumber": "1770000000",
    "LifetimeData": {
      "compressions": -1,
      "ventilations": -1
    }
  },
  "Debriefings": [
    {
      "text": "improvement-need-hand-position"
    },
    {
      "text": "tip-comp-hand-position"
    }
  ],
  "SimulationActivity": {
    "productId": "HeartCode BLS",
    "productVersion": "4.1.0.20918786131",
    "simulationVersion": "1.0.0.0",
    "brandingId": "AHA",
    "scoId": "skills-web-adult-compressions",
    "SkillsScenarioType": {
      "name": "Adult Compressions",
      "baseline": false,

```

```

        "sessionType": "Assessment",
        "analyticsActivity": "Adult Compressions (HeartCode
BLS)",
        "sessionOutcome": "completed",
        "Requires": [
            {
                "type": "Compressions",
                "completed": true,
                "count": 60
            }
        ],
        "sessionTimeLimit": 120
    }
}
},
"course_id": "manifest/skills_2025_hc_bls_dev",
"activities": [
    "hc-overview",
    "skills-web-adult-compressions",
    "skills-web-adult-ventilations",
    "skills-web-adult-2CPR-30-2",
    "skills-web-infant-compressions",
    "skills-web-infant-ventilations",
    "skills-web-infant-2CPR-15-2"
],
"course_progress": {
    "hc-overview": {
        "_score": 0,
        "_complete": 1,
        "_elapsed": 20663,
        "_failures": 0
    },
    "skills-web-adult-compressions": {
        "debriefingScore": 8,
        "passedCourse": 0,
        "completedScenario": 1,
        "_score": 0.08,
        "_complete": 0,
        "_elapsed": 60730,
        "_failures": 1
    }
}
}
}
}

```

Payload Fields

Top-Level Fields

Field	Type	Description
<code>requestId</code>	string	Unique request identifier for idempotency
<code>eventType</code>	string	Always <code>skills-activity-ended</code>
<code>timestamp</code>	string	ISO 8601 UTC timestamp when event occurred
<code>stationId</code>	string	Fleet Management station identifier
<code>data</code>	object	Event-specific payload (see below)

Data Object Fields

Field	Type	Description
<code>launch_id</code>	string	Launch identifier for deduplication of delivery
<code>org_id</code>	string	RQI1Stop organization identifier
<code>manikin_id</code>	string	Manikin serial number
<code>user_email</code>	string	Learner's email address
<code>user_id</code>	string	Learner's RQI1Stop user identifier
<code>user_name</code>	string	Learner's full name
<code>activity_id</code>	string	Activity/SCO identifier
<code>activity_completed</code>	boolean	Indicates whether the learner met all completion criteria
<code>activity_score</code>	number	Final score for the attempt (0.0 – 1.0)
<code>activity_elapsed</code>	number	Total milliseconds elapsed in the attempt, including pauses and idle time
<code>activity_failures</code>	number	Total failed attempts for this activity within the session
<code>organization_name</code>	string	Organization display name
<code>institution_id</code>	string	Institution identifier
<code>institution_name</code>	string	Institution display name
<code>time_zone</code>	string	Learner's time zone (for example, <code>Europe/Dublin</code>)

<code>activity_result</code>	object	SSJ payload with detailed performance metrics
<code>course_id</code>	string	Course/manifest identifier
<code>activities</code>	array	Ordered list of activity IDs in the course
<code>course_progress</code>	object	Updated per-activity progress summary for the course

`course_progress` uses activity IDs as keys. Each entry exposes `_score`, `_complete`, `_elapsed`, `_failures`, and activity-specific metrics (for example `completedScenario`) that reflect the learner's history up to the close of the attempt.

Activity Result Object

The `activity_result` payload captures detailed performance data generated by Skills Software. Key sections include:

- `SSJVersion` – Schema version used for the SSJ (Skills Session JSON) payload
- `CprSessionStatistics` – Raw CPR performance metrics
- `CprSessionScores` – Scoring and assessment breakdown
- `DeviceInfo` – Manikin hardware identifiers and lifetime counters
- `Debriefings` – Feedback messages shown to the learner
- `SimulationActivity` – Scenario metadata, requirements, and outcomes

CprSessionStatistics fields

- `compCount` – Total number of compressions
- `compCyclesCount` – Total number of compression sets
- `ventCount` – Total number of ventilations
- `ventCyclesCount` – Total number of ventilation sets

- `compInactivityPeriodCount` – Number of compression inactivity periods
- `compMeanDepth` – Mean compression depth (null if unavailable)
- `compStdDeviationDepth` – Standard deviation of compression depth (null if unavailable)
- `compMeanReleaseDepth` – Mean compression release depth (null if unavailable)
- `compStdDeviationReleaseDepth` – Standard deviation of release depth (null if unavailable)
- `compMeanRate` – Mean compression rate (null if unavailable)
- `compStdDeviationRate` – Standard deviation of compression rate (null if unavailable)
- `compCorrectDepthPercentage` – Percentage of compressions with correct depth
- `compCorrectReleasePercentage` – Percentage of compressions with full release
- `compCorrectRatePercentage` – Percentage of compressions within the correct rate band
- `compTooLowRatePercentage` – Percentage of compressions below the target rate
- `compHandPosCorrectPercentage` – Percentage of compressions with correct hand position
- `inactivityAverageTime` – Average duration of inactivity periods (seconds)
- `meanVentilationInflationTime` – Mean inflation time in milliseconds (optional)
- `stdDevVentilationInflationTime` – Standard deviation of inflation time (optional)
- `numPreVents` – Number of pre-ventilations recorded (optional)
- `averageNumberOfVentilationsPerCycle` – Mean ventilations per cycle (optional)
- `minNumberOfVentilationsPerCycle` – Minimum ventilations in a cycle (optional)
- `maxNumberOfVentilationsPerCycle` – Maximum ventilations in a cycle (optional)

- `averageNumberOfCompressionsPerCycle` – Mean compressions per cycle (optional)
- `minNumberOfCompressionsPerCycle` – Minimum compressions in a cycle (optional)
- `maxNumberOfCompressionsPerCycle` – Maximum compressions in a cycle (optional)
- `ventMeanVolume` – Mean ventilation volume (optional)
- `ventStdDeviationVolume` – Standard deviation of ventilation volume (optional)
- `ventMeanRate` – Mean ventilation rate (optional)
- `ventStdDeviationRate` – Standard deviation of ventilation rate (optional)
- `ventCorrectVolumePercentage` – Percentage of ventilations with correct volume (optional)
- `ventNotEnoughVolumePercentage` – Percentage of ventilations with insufficient volume (optional)
- `ventTooMuchVolumePercentage` – Percentage of ventilations with excessive volume (optional)
- `ventCorrectRatePercentage` – Percentage of ventilations at correct rate (optional)
- `ventTooLowRatePercentage` – Percentage of ventilations below target rate (optional)
- `ventCorrectInflationTimePercentage` – Percentage of ventilations with correct inflation time (optional)
- `ventTooFastInflationTimePercentage` – Percentage of ventilations with inflation time that was too short (optional)
- `cprCyclesAccepted` – Count of CPR cycles that met the acceptance criteria (optional)
- `flowTimePercentage` – Percentage of session time with active compressions

CprSessionScores fields

- `cprScore` – CPR score in percent
- `overallScore` – Overall score in percent

- `categoryThresholdBasic` – Threshold for passing as “basic”
- `categoryThresholdAdvanced` – Threshold for passing as “advanced”
- `compReleaseScoreReduction` – Score reduction for incomplete release (optional)
- `compDepthShallowScoreReduction` – Score reduction for shallow compressions (optional)
- `compRateLowScoreReduction` – Score reduction for too-slow compressions (optional)
- `compRateHighScoreReduction` – Score reduction for too-fast compressions (optional)
- `compHandposScoreReduction` – Score reduction for incorrect hand position (optional)
- `compOverallScore` – Compressions score in percent (optional)
- `ventShallowScoreReduction` – Score reduction for low ventilation volume (optional)
- `ventExcessiveScoreReduction` – Score reduction for excessive ventilation volume (optional)
- `ventRateLowScoreReduction` – Score reduction for slow ventilation rate (optional)
- `ventRateHighScoreReduction` – Score reduction for fast ventilation rate (optional)
- `ventCyclesScoreReduction` – Score reduction for incorrect number of ventilations per cycle (optional)
- `ventOverallScore` – Ventilation score in percent (optional)
- `compDeepDepthScoreReduction` – Score reduction for overly deep compressions (optional)
- `compCyclesScoreReduction` – Score reduction for incorrect compression counts (optional)
- `compFlowtimeScoreReduction` – Score reduction for low compression fraction (optional)
- `preVentsTooFewScoreReduction` – Score reduction for too few pre-ventilations (optional)

- `preVentsTooManyScoreReduction` – Score reduction for too many pre-ventilations (optional)
- `inflationTimeFastScoreReduction` – Score reduction for fast ventilations (optional)
- `inflationTimeSlowScoreReduction` – Score reduction for slow ventilations (optional)
- `isScoringModeAED` – Indicates whether AED scoring mode was used
- `compFlowTimeScore` – Compression flowtime score (optional)
- `compDepthScore` – Compression depth score (optional)
- `compRateScore` – Compression rate score (optional)
- `compReleaseScore` – Compression release score (optional)
- `compHandPositionScore` – Hand position score (optional)
- `ventRateScore` – Ventilation rate score (optional)
- `ventVolumeScore` – Ventilation volume score (optional)
- `aedScore` – Overall AED score (optional)
- `aedAppliedPadsScore` – AED pad application score (optional)
- `aedShockDeliveredScore` – AED shock delivery score (optional)
- `aedCprStartedScore` – Score for resuming CPR after AED prompts (optional)
- `aedCprAfterNoShockScore` – Score for resuming CPR after a no-shock decision (optional)
- `virtualRescuerInterferenceScoreReduction` – Score reduction for interfering with the virtual rescuer (optional)
- `aedInterferenceScoreReduction` – Score reduction for interfering during AED analysis (optional)
- `aedMissingAnalysisStayClearScoreReduction` – Score reduction for missing “clear” before analysis (optional)
- `aedMissingShockStayClearScoreReduction` – Score reduction for missing “clear” before shock (optional)
- `scanChestTimeSlowScoreReduction` – Score reduction for scanning too slowly (optional)
- `scanChestTimeFastScoreReduction` – Score reduction for scanning too quickly (optional)

- `pipScore` – Peak inspiratory pressure score (optional)
- `maskSealScore` – Mask seal score (optional)
- `headPositionScore` – Head position score (optional)

DeviceInfo

- `serialNumber` – Manikin serial number
- `LifetimeData.compressions` – Lifetime compression count
- `LifetimeData.ventilations` – Lifetime ventilation count

Debriefings

Entries in `Debriefings` describe why the learner failed or what to improve:

Improvements

- `improvement-comp-set-size-high` – Too many compressions in some cycles
- `improvement-comp-set-size-incorrect` – Incorrect compression counts detected
- `improvement-comp-set-size-low` – Not enough compressions in some cycles
- `improvement-need-aed-interruptions` – Do not interrupt the AED sequence
- `improvement-need-compression-rate` – Compression rate needs improvement
- `improvement-need-depth` – Compression depth needs improvement
- `improvement-need-detectable-compressions` – Perform at least the minimum number of compressions
- `improvement-need-detectable-compressions-time` – Maintain compressions for the required duration
- `improvement-need-detectable-ventilations` – Perform at least the minimum number of ventilations
- `improvement-need-detectable-ventilations-time` – Maintain ventilations for the required duration

- `improvement-need-finger-position` – Finger placement needs improvement
- `improvement-need-flowtime` – Improve chest compression fraction
- `improvement-need-hand-position` – Hand placement needs improvement
- `improvement-need-none-text` – Learner scored 100%
- `improvement-need-none-title` – Congratulatory message for perfect performance
- `improvement-need-prevents-number` – Incorrect number of pre-ventilations
- `improvement-need-release` – Compression release needs improvement
- `improvement-need-ventilation-rate` – Ventilation rate needs improvement
- `improvement-need-ventilation-volume` – Ventilation volume needs improvement
- `improvement-need-virtual-rescuer-interruptions` – Do not interrupt the co-rescuer
- `improvement-scenario-not-completed` – Not all required actions were performed
- `improvement-set-size-incorrect` – Set sizes were incorrect in some cycles
- `improvement-vent-set-size-high` – Too many ventilations in some cycles
- `improvement-vent-set-size-incorrect` – Incorrect ventilation counts detected
- `improvement-vent-set-size-low` – Not enough ventilations in some cycles

Tips

- `tip-aed-early-failure` – Incorrect AED sequence
- `tip-aed-early-failure-1` – Reminder to turn on the AED
- `tip-aed-early-failure-2` – Follow AED prompts precisely
- `tip-aed-timeout-after-no-shock-advised` – Resume CPR promptly after a no-shock decision
- `tip-aed-timeout-compressions` – Resume compressions promptly after shock
- `tip-aed-timeout-cpr` – Resume CPR promptly after shock
- `tip-aed-timeout-pads` – Attach AED pads within the allowed time

- `tip-aed-timeout-shock` – Deliver the shock within the allowed time
- `tip-comp-depth-incorrect` – Compression depth inconsistent
- `tip-comp-depth-shallow` – Compressions too shallow
- `tip-comp-depth-too-deep` – Compressions too deep
- `tip-comp-finger-position` – Correct finger placement reminder
- `tip-comp-hand-position` – Correct hand placement reminder
- `tip-comp-minimize-interruptions` – Minimize interruptions to maintain perfusion
- `tip-comp-rate-high` – Compression rate too high
- `tip-comp-rate-high-and-low` – Compression rate inconsistent
- `tip-comp-rate-incorrect` – Compression rate incorrect
- `tip-comp-rate-low` – Compression rate too low
- `tip-comp-release` – Release pressure fully between compressions
- `tip-comp-too-few` – Not enough compressions completed in time
- `tip-duration-too-less` – Required action not performed for long enough
- `tip-no-activity-detected` – No manikin activity detected
- `tip-pause-detected` – Pause detected for a specified duration
- `tip-scenario-not-completed` – Required actions not completed
- `tip-vent-rate` – Ventilation rate incorrect
- `tip-vent-rate-high` – Ventilation rate too high
- `tip-vent-rate-high-and-low` – Ventilation rate inconsistent
- `tip-vent-rate-incorrect` – Ventilation rate incorrect
- `tip-vent-rate-low` – Ventilation rate too low
- `tip-vent-too-few` – Not enough ventilations completed in time
- `tip-vent-volume` – Ventilation volume incorrect
- `tip-vent-volume-excessive` – Ventilation volume too high
- `tip-vent-volume-incorrect` – Ventilation volume inconsistent
- `tip-vent-volume-shallow` – Ventilations too shallow

SimulationActivity

- `productId` – Identifier for the product (for example, `HeartCode BLS`)
- `productVersion` – Product version string
- `simulationVersion` – Simulation engine version
- `brandingId` – Branding identifier (for example, `AHA`)
- `scoId` – SCO identifier for the course
- `SkillsScenarioType` – Extended scenario metadata with:
 - `name` – Descriptive scenario name
 - `baseline` – Indicates whether this was a baseline assessment
 - `sessionType` – Scenario type (`Baseline` , `Warmup` , `HelpLane` , `Assessment`)
 - `sessionTimeLimit` – Time limit in seconds (optional)
 - `analyticsActivity` – Course and session label used for analytics bucketing
 - `switchesPositions` – Indicates whether position switching is required (optional)
 - `sessionOutcome` – Describes how the session ended (see table below)
 - `Requires` – Ordered list of requirements with `type` , `completed` , and optional `count` , `duration` , `timeout` , `scored` , `mustPassAllCycles` , and `scoredCount`

sessionOutcome values

Value	Meaning
<code>completed</code>	Session completed within the time limit
<code>session-time-out</code>	Session exceeded the time limit
<code>no-manikin-activity</code>	No manikin activity was registered
<code>early-fail-no-aed</code>	Learner skipped required AED steps before proceeding
<code>early-fail-cpr</code>	Learner performed the wrong CPR sequence for the scenario
<code>partial-manikin-activity</code>	Session timed out with only partial manikin activity recorded

The `analyticsActivity` string follows the format `<courseName> (<productName>)`, where `productName` may be `HeartCode ACLS`, `HeartCode BLS`, or `HeartCode PALS`.

Activity Lifecycle Notes

- `activity_completed` distinguishes between successful completion (`true`) and attempts that ended early or failed (`false`).
- `activity_elapsed` includes all time spent, even if the learner pauses; use `activity_result.SimulationActivity.sessionOutcome` to understand why the attempt ended.
- Learners who close the browser or lose connection before completion will not produce this event, so reconcile progress based on what you receive rather than assuming a matching end event.
- When a learner remains idle, the platform times out the attempt and sends this event with a timeout outcome, ensuring downstream systems can close the session.
- `activity_failures` counts failed attempts within the session; pair it with `skills-activity-started` events to track repeated retries.

Use Cases

- Record training completion and scores
- Analyze learner performance metrics
- Trigger certificates or notifications on completion
- Identify learners who need additional support (low scores, multiple failures)
- Generate training reports and analytics
- Track course progress and completion rates

Example Implementation

```
app.post('/webhooks/fleet-management', async (req, res) => {
  // Verify signature (see Webhooks Overview)
  if (!verifySignature(req)) {
    return res.status(401).send('Invalid signature');
  }

  const event = req.body;

  if (event.eventType === 'skills-activity-ended') {
    const {
      user_email,
      user_name,
      activity_id,
      activity_completed,
      activity_score,
      activity_failures,
      activity_result
    } = event.data;

    // Process activity completion
    await recordActivityCompletion({
      userEmail: user_email,
      userName: user_name,
      activityId: activity_id,
      completed: activity_completed,
      score: activity_score,
      failures: activity_failures,
      detailedResults: activity_result,
      timestamp: event.timestamp
    });

    // Trigger follow-up actions based on performance
    if (activity_failures > 2) {
      await notifyInstructor({
        userEmail: user_email,
        activityId: activity_id,
        reason: 'Multiple failures detected'
      });
    }

    // Respond immediately
  }
});
```

```
res.status(200).send('OK');  
}  
});
```

Related Events

- [skills-activity-started](#) - Triggered when the activity begins

Previous
[skills-activity-started](#)

Next
[Analytics](#)

Last updated 5 months ago